

$$H = \tfrac{1}{2}L_s\|\mathbf{e}_s\|^2 + \tfrac{1}{2}K_i\|\mathbf{z}_s\|^2, \quad \dot{H} = -(R_s + R_a)\|\mathbf{e}_s\|^2 \leq 0$$

workless coupling $\omega_e L_s\boldsymbol{\mathcal{J}}$ kept ($\mathbf{x}^T\boldsymbol{\mathcal{J}}\mathbf{x} = 0$)